

# fixed

## Fixed

**Revision:** 9 **Last modified:** 2026-06-01T11:05:59Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db\_to\_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: 77.

| HXC      | Type    | Pri | Title                                                                     | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------|---------|
| HXC-001  | Task    | P1  |                                                                           |         |
| HXC-002  | Task    | P1  |                                                                           |         |
| HXC-003  | Task    | P1  |                                                                           |         |
| HXC-004  | Task    | P1  |                                                                           |         |
| HXC-005  | Task    | P1  |                                                                           |         |
| HXC-006  | Task    | P1  |                                                                           |         |
| HXC-007  | Task    | P1  |                                                                           |         |
| HXC-008  | Task    | P1  |                                                                           |         |
| HXC-009  | Task    | P1  |                                                                           |         |
| HXC-010  | Task    | P1  |                                                                           |         |
| HXC-011  | Task    | P1  |                                                                           |         |
| HXC-012  | Task    | P1  |                                                                           |         |
| HXC-013  | Task    | P1  |                                                                           |         |
| HXC-014  | Task    | P1  |                                                                           |         |
| HXC-015  | Task    | P1  |                                                                           |         |
| HXC-1001 | Task    | P0  | Stand up embedded-etcd integration harness for pkg/etcd tests             | dbed10c |
| HXC-1002 | Feature | P0  | Prove etcd Put/Get round-trip and missing-key semantics against real etcd | dbed10c |
| HXC-1003 | Feature | P1  | Prove etcd GetPrefix returns exactly matching multi-key set               | dbed10c |
| HXC-1004 | Feature | P1  | Prove etcd Delete/DeletePrefix remove keys (sink-side absence)            | dbed10c |
| HXC-1005 | Feature | P0  | Prove etcd Watch fires correct event and closes on ctx cancel             | dbed10c |

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|----------|---------|-----|-------------------------------------------------------------------------------------------------|---------|
| HXC-1006 | Feature | P1  | Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior                                | dbed10c |
| HXC-1007 | Feature | P0  | Prove etcd distributed Lock mutual exclusion (highest-risk path)                                | dbed10c |
| HXC-1008 | Task    | P2  | Add mutation-paired etcd New() default-timeout injection test                                   | dbed10c |
| HXC-1009 | Task    | P2  | Prove etcd error-wrapping on dead endpoint                                                      | dbed10c |
| HXC-1016 | Feature | P0  | Add EtcdLocker integration tests against real etcd (block/acquire/lease)                        | dbed10c |
| HXC-1017 | Bug     | P0  | Rewrite TestMemoryLockerConcurrent to detect a broken lock without -race                        | dbed10c |
| HXC-1018 | Task    | P2  | Add explicit block-then-acquire ordering test for MemoryLocker                                  | dbed10c |
| HXC-1021 | Bug     | P1  | Drive pkg/build StateFailed via a genuine build failure not the 'fail' sentinel                 | dbed10c |
| HXC-1024 | Bug     | P2  | Enforce or document pkg/build content-addressable cache digest integrity                        | dbed10c |
| HXC-1026 | Feature | P0  | Add real mTLS handshake test for pkg/security (valid client succeeds, no/foreign cert rejected) | dbed10c |
| HXC-1027 | Bug     | P1  | Assert MinVersion==TLS13 on the client TLS path in pkg/security                                 | dbed10c |
| HXC-1030 | Bug     | P1  | Verify pkg/hxcregistry item_history audit-sink rows after Create/Update                         | dbed10c |
| HXC-1031 | Bug     | P2  | Assert exact and stable ComputeHeadingHash value in pkg/hxcregistry                             | dbed10c |
| HXC-1032 | Task    | P2  | Add pkg/hxcregistry failure-path tests (dup PK, missing-id Get/Update)                          | dbed10c |
| HXC-1033 | Bug     | P2  | Make pkg/hxcregistry concurrent test do concurrent writes (or rename)                           | dbed10c |
| HXC-1034 | Task    | P3  | Prove pkg/hxcregistry timestamp round-trip and DB CHECK-constraint enforcement                  | dbed10c |
| HXC-1035 | Bug     | P0  | Wire pkg/wasm WasmPlugin.Execute to real WASM output and prove transformed result               | dbed10c |
| HXC-1036 | Task    | P1  | Add pkg/wasm negative tests for missing execute export and corrupt module Init                  | dbed10c |
| HXC-1037 | Bug     | P2  | Prove pkg/wasm init/shutdown exports actually run via memory side-effect                        | dbed10c |
| HXC-1039 | Bug     | P1  | Replace self-defeating pkg/semaphore TestSemaphore_ReleaseTooMany_Mutation                      | a084eb1 |
| HXC-1040 | Feature | P1  | Add pkg/semaphore M>>N concurrency invariant test under -race                                   | a084eb1 |
| HXC-1041 | Task    | P3  |                                                                                                 | a084eb1 |

| HXC      | Type    | Pri | Title                                                                                       | Commit  |
|----------|---------|-----|---------------------------------------------------------------------------------------------|---------|
|          |         |     | Add pkg/semaphore New(0) barrier and exact-capacity boundary tests                          |         |
| HXC-1042 | Bug     | P1  | Convert pkg/storage error tests to errors.Is on exported sentinels                          | dbed10c |
| HXC-1043 | Bug     | P2  | Reconcile and test pkg/storage FileStore vs MemoryStore copy semantics                      | dbed10c |
| HXC-1044 | Task    | P2  | Add pkg/storage overwrite, missing-delete, and List edge tests for both stores              | dbed10c |
| HXC-1045 | Feature | P1  | Add pkg/storage -race concurrent Put/Get/Delete test to prove RWMutex                       | dbed10c |
| HXC-1046 | Bug     | P1  | Gate -race in CI for pkg/metrics concurrency mutation tests                                 | a084eb1 |
| HXC-1047 | Bug     | P2  | Complete pkg/metrics Prometheus exposition assertions (+Inf, _sum, ordering)                | a084eb1 |
| HXC-1048 | Task    | P2  | Add pkg/metrics duplicate-registration and histogram edge tests                             | a084eb1 |
| HXC-1049 | Task    | P3  | Add pkg/metrics /metrics integration test via httptest.Server                               | a084eb1 |
| HXC-1050 | Bug     | P1  | Test pkg/grpcutil LoggingStreamInterceptor recv/send counters                               | a084eb1 |
| HXC-1051 | Bug     | P2  | Strengthen pkg/grpcutil log assertions to dynamic behavior-dependent fields                 | a084eb1 |
| HXC-1052 | Task    | P2  | Cover pkg/grpcutil status.FromError code extraction for handler errors                      | a084eb1 |
| HXC-1053 | Bug     | P1  | Add pkg/context<br>TestDetach_PreservesValues for the value-retention guarantee             | a084eb1 |
| HXC-1054 | Bug     | P2  | Strengthen pkg/context WithTimeout deadline and expiry assertions                           | a084eb1 |
| HXC-1055 | Bug     | P1  | Give pkg/workerpool<br>TestPool_NilWorkSafe_Mutation a real sink                            | a084eb1 |
| HXC-1056 | Feature | P1  | Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recovery, idempotent-Stop tests | a084eb1 |
| HXC-1057 | Bug     | P1  | Make pkg/errors concurrency test honest via locked GetFields path + -race                   | a084eb1 |
| HXC-1058 | Task    | P2  | Pin pkg/errors Code enum exact values, distinctness, and WithFields nil receiver            | a084eb1 |
| HXC-1059 | Bug     | P2  | Replace tautological pkg/backoff cap-mutation test and pin Default()                        | a084eb1 |
| HXC-1060 | Bug     | P2  | Assert pkg/classads parser AST Op per operator and paren precedence                         | a084eb1 |
| HXC-1061 | Task    | P3  | Add pkg/classads eval failure-path and regexp non-match tests                               | a084eb1 |
| HXC-1062 | Bug     | P2  |                                                                                             | dbed10c |

| HXC      | Type    | Pri | Title                                                                                      | Commit  |
|----------|---------|-----|--------------------------------------------------------------------------------------------|---------|
|          |         |     | Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is checks                  |         |
| HXC-1063 | Task    | P3  | Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBKDF2 vector tests       | dbed10c |
| HXC-1064 | Bug     | P1  | Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug                       | a084eb1 |
| HXC-1065 | Task    | P3  | Strengthen pkg/log Warn test, add Debug-suppression, remove dead captureOutput scaffolding | a084eb1 |
| HXC-1066 | Task    | P3  | Add pkg/lru boundary, cold-miss, and struct-value round-trip tests                         | a084eb1 |
| HXC-1067 | Feature | P2  | Add pkg/pubsub -race concurrency test and explicit drop-boundary assertion                 | a084eb1 |
| HXC-1068 | Bug     | P2  | Prove pkg/ratelimit bucket starts full and add Limiter conformance + -race tests           | a084eb1 |
| HXC-1069 | Bug     | P1  | Bring pkg/retry DoWithResult to behavioral parity with Do                                  | a084eb1 |
| HXC-1070 | Task    | P2  | Add pkg/retry transient-recovery, jitter-bound, and computeDelay table tests               | a084eb1 |
| HXC-1071 | Task    | P3  | Strengthen pkg/serde MustMarshal exact-content and add Unmarshal failure tests             | a084eb1 |
| HXC-1072 | Bug     | P2  | Pin pkg/validator error contract via errors.Is and remove/wire unused sentinels            | a084eb1 |
| HXC-1073 | Task    | P3  | Add pkg/validator uint min/max, IsValidID empty, oneof/email boundary tests                | a084eb1 |
| HXC-1074 | Task    | P0  | Establish fleet-wide -race CI gate across concurrency-sensitive packages                   | a084eb1 |
| HXC-902  | Task    | P1  |                                                                                            |         |
| HXC-903  | Task    | P1  |                                                                                            |         |